

[Swift](#) / ...

Protocol

CVarArg

A type whose instances can be encoded, and appropriately passed, as elements of a C `va_list`.

iOS 8.0+ | iPadOS 8.0+ | Mac Catalyst 13.0+ | macOS 10.10+ | tvOS 9.0+ | visionOS 1.0+ | watchOS 2.0+

`protocol CVarArg`

Mentioned in

 [Using Imported C Functions in Swift](#)

Overview

You use this protocol to present a native Swift interface to a C “varargs” API. For example, a program can import a C API like the one defined here:

```
int c_api(int, va_list arguments)
```

To create a wrapper for the `c_api` function, write a function that takes `CVarArg` arguments, and then call the imported C function using the `withVaList(_:_:_)` function:

```
func swiftAPI(_ x: Int, arguments: CVarArg...) -> Int {
    return withVaList(arguments) { c_api(x, $0) }
}
```

Swift only imports C variadic functions that use a `va_list` for their arguments. C functions that use the `...` syntax for variadic arguments are not imported, and therefore can't be called using